



***ClassCube*: Effective and Efficient Big OLAP Data Cube Classification via Dimensionality Reduction**

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Abstract. In this paper, we introduce *ClassCube*, a novel methodology designed to perform *efficient* and *effective* classification over large, multidimensional OLAP data cubes. *ClassCube* leverages *logical cuboid lattices* to represent data across multiple aggregation levels, enabling intelligent selection of both dimensions and cuboids for classification tasks. By integrating dimensionality reduction techniques such as *Dimension Selection* and *Principal Component Analysis*, the approach significantly reduces computational overhead while maintaining high classification accuracy. Extensive experimental assessments confirm that *ClassCube* achieves an optimal balance between *efficiency* and *accuracy*, highlighting its suitability for *real-life big data analytics applications*.

Keywords: Big Data Analytics · Multidimensional Big Data Analytics · Integration of OLAP Analysis and Classification

1 Introduction

In today's data-driven era, organizations across various sectors, such as finance, healthcare, and e-commerce, rely heavily on analyzing large, multidimensional datasets to make strategic decisions. In this context, *Online Analytical Processing* (OLAP) data cubes have become fundamental tools, enabling complex queries and interactive exploration of data aggregate over multiple dimensions. These data cubes facilitate operations like slicing, dicing, and drilling down into data, which are essential for uncovering patterns and trends that lead to business insights.

However, as the volume and dimensionality of data continue to grow exponentially, performing classification tasks directly on OLAP data cubes has become increasingly computationally expensive. High-dimensional data presents significant challenges, especially the *curse of dimensionality*, where the feature space becomes so huge that data

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points become sparse. This sparsity adversely affects the performance of classification algorithms, leading to longer computation times and potential overfitting. Consequently, there is a need for efficient techniques that can reduce computational costs while maintaining high performance of classification.

The main challenge focused on in this research regards the substantial computational load associated with performing classification on high-dimensional OLAP data cubes. Traditional classification methods struggle with scalability in such areas due to the extensive resources required for processing and the potential degradation in accuracy caused by the high number of dimensions. This issue is particularly acute in real-time and mobile environments, where computational resources are limited. Therefore, enhancing classification processes to handle high-dimensional data efficiently is crucial for enhancing the effectiveness of data analysis in various applications.

Several research efforts have focused on addressing the computational challenges associated with high-dimensional data classification, particularly in the context of large datasets. *Dimensionality reduction techniques* (e.g., [1]), such as *Principal Component Analysis* (PCA) [2] and *Feature Selection Algorithms* (e.g., [3]), have been widely employed to mitigate the curse of dimensionality. For instance, [4] integrates PCA into a multidimensional F1-transform classifier, effectively reducing computational loads and improving classification accuracy over traditional algorithms. Similarly, [5] develops a *hybrid optimization-based feature selection method* to identify relevant feature subsets, resulting in improved convergence speed and classification performance on high-dimensional datasets. Beyond using dimensionality reduction methods, some studies have explored the integration of these techniques with classification algorithms to further enhance efficiency. [6] introduces *deep self-supervised machine learning* models enriched with novel feature elimination and selection strategies, effectively reducing dimensionality and improving classification accuracy for multidimensional health risk assessments. Additionally, advanced data structuring methods have been proposed to manage high-dimensional data more effectively. [7] presents an *adaptive granularity* and *dimension decoupling network* for multidimensional time series classification, which extracts features at various scales and decouples dimensions to prevent dominant features from overshadowing others.

To tackle the aforementioned challenges, we propose *ClassCube*, a novel methodology that integrates *multidimensional OLAP analysis* [8] and *classification algorithms* [9] to effectively and efficiently support *big data analytics* in real-life scenarios. The key idea of *ClassCube* relies on dimensionality reduction tools (e.g., [1]). At a practical level, we leverage on so-called *big OLAP data cubes* [10] for big data applications, and we address the issue of *effectively and efficiently classifying big multidimensional data* [11] *in Cloud environments* (e.g., [12]). With this goal in mind, we propose the anatomy and main functionalities, including experimental assessment and evaluation, of *ClassCube*, *an innovative methodology for supporting advanced big data analytics via intelligent classification tools over big OLAP data cubes*.

In our study, we focus on leveraging the *logical cuboid lattice*, a hierarchical structure that represents all possible aggregations of the data across different combinations of dimensions. Each cuboid in the lattice corresponds to a specific aggregation level, offering a multi-resolution view of the data. This structure leads the model to more efficient

data management and analysis by enabling operations at different levels of granularity. However, even with this hierarchical approach, performing classification directly on the entire lattice remains resource-intensive. It should be noted that the selection process focuses on identifying specific dimensions from the OLAP data cube to define the cuboids of interest. This criterion can be guided by *user/application input* or determined based on *state-of-the-art models* (e.g., [13, 14]).

Our proposed methodology integrates dimensionality reduction with hierarchical data cube structuring to perform an efficient classification process. Specifically, we extract the *logical cuboid lattice* from the original OLAP data cube using *Principal Component Analysis* to construct a new, reduced-dimensionality data cube. PCA effectively reduces the number of dimensions by projecting the most significant features that contribute to data variance, to a reduced feature space. Thus, mitigating the curse of dimensionality. On the other hand, we apply a dimension selection method to extract the reduced-dimensionality data cubes, which then the classification performance is compared with the one resulting from PCA utilization. Thus, the reduced data cube serves as the basis for our classification tasks.

We then perform classification using algorithms such as *Logistic Regression* (LR) and *Support Vector Machines* (SVM) on the cuboids of interest. The reduced dimensionality substantially lowers computational costs while aiming to preserve classification accuracy. Evaluating the effectiveness of this approach presents its own challenges. Direct comparison with the original high-dimensional data cube is impractical due to resource constraints. Therefore, we compare the classification performance on the reduced data cube with that on cuboids from the lattice that have the same dimensionality. This comparative strategy provides a meaningful evaluation for assessing the effectiveness of our methodology.

A preliminary, introductive version of this research appears in [31]. The key contributions of this research are as follows:

- First, we introduce an efficient dimensionality reduction framework that employs a hierarchical data cube structure to be used in classification applications on high-dimensional OLAP data cubes.
- Second, our methodology applies classification algorithms on the cuboids of interest from the extracted *lattice of cuboids*.
- Third, we establish a practical evaluation methodology by comparing the reduced data cubes (i.e., cuboids) from the lattice, providing meaningful insights into the effectiveness of our approach.

2 Related Work

In this Section, we review relevant related work that aligns with our objective of efficient and effective classification of multidimensional OLAP data cubes through the integration of dimensionality reduction techniques.

PCA [2] is one of the most widely used *dimensionality reduction methods*, capable of preserving the most significant variance components. In [4], it is used to enhance the multidimensional F1-transform classifier by reducing data size during preprocessing, which lowers computational costs. Their hybrid approach improves classification

accuracy and outperforms traditional classifiers such as *SVM*, *Random Forests* (RF), and *Artificial Neural Networks* (ANN), which demonstrates the effectiveness of PCA in high-dimensional spaces.

Similarly, [5] proposes a *hybrid optimization-based feature selection method* by combining *Slime Mould Algorithm* (SMA) and *Binary Grey Wolf Optimization* (BGWO) to improve classification performance while reducing computational loads.

This approach efficiently identifies relevant features from high-dimensional data, mitigating the curse of dimensionality, and improves *K-Nearest Neighbor* (KNN) classifier results over SVM and Naïve Bayes. [15] employs a *multi-objective genetic algorithm* (GA) guided by feature importance rankings to optimize both accuracy and dimensionality reduction. Their method achieves *Pareto-optimal solutions* and consistently outperforms traditional feature selection techniques across diverse high-dimensional datasets.

Furthermore, the integration of dimensionality reduction techniques with classification algorithms has proven to be an effective strategy in improving performance on high-dimensional datasets. [6] tackles the classification of multidimensional health risks from blood test data using *deep self-supervised learning* enhanced by novel *feature elimination* and *weighting techniques*. They introduce an *adaptive feature elimination method* to reduce redundant data and a *self-feature weighting approach* that prioritizes features deviating from normal values. These enhancements, applied to algorithms like *Batch Least Squares* (BLS) and *Iterative Neural Networks* (INN), improve classification accuracy while reducing computational complexity. On the other hand, [16] addresses the computational challenges of applying *Shapley values* for high-dimensional, time-series data in the context of DNA profile classification using *convolutional neural networks* (CNNs). To overcome this issue, they introduce a novel *partitioning strategy* suitable for time-series data inspired by superpixels in image processing. By segmenting time-series data into meaningful blocks, they enable efficient Shapley value computation without incurring high computational costs.

In the area of *multidimensional functional data*, [17] proposes a *multiclass functional deep neural network* (mfDNN) that combines Functional Data Analysis with deep learning. Using *Functional Principal Component Analysis* (FPCA) for dimensionality reduction, they extract *principal component scores* and input them into a sparse deep neural network with ReLU activation. The mfDNN classifier shows strong classification performance on both simulated and real-life datasets like MNIST and *Alzheimer's Disease Neuroimaging Initiative* (ADNI). [18] presents a deep learning method for classifying epileptic seizures from high-dimensional *Electroencephalography* (EEG) signals. They convert 1D-EEG signals into 2D and 3D spectrograms using *Short-Time Fourier Transform* (STFT) to capture *time-frequency features*. A *deep convolutional autoencoder* (DCAE) then extracts hierarchical features from these spectrograms, enabling effective seizure classification.

In the context of *multidimensional classification* (MDC), [19] introduces a *probability-based label enhancement method* for MDC, where instances belong to multiple heterogeneous class spaces. To address the limitations of traditional methods, they encode class spaces into a *single logical-label space* and minimize the *Kullback-Leibler divergence* between feature and label distribution manifolds. [20] tackles the *Imbalance*

Shift problem in MDC, where class importance varies across labeling dimensions. They propose the *Imbalance-Aware Fusion Model* (IMAM), which decomposes the MDC task into separate *multi-class problems*. Each is handled by an *imbalance-aware deep model* that adjusts prediction logits based on class distributions.

Moreover, *Feature Fusion methods* have been also used in research to mitigate dimensional complexity. [21] addresses the challenge of accurately classifying multiple respiratory diseases by combining *one-dimensional* and *two-dimensional* data representations. By employing widely used 2D representations like *spectrograms* and *mel-frequency cepstral coefficients* (MFCC) alongside raw 1D respiratory data, they create a dataset that includes both *time-series* and *frequency-domain features*. [22] proposes a method that combines multidimensional feature fusion of *Long Short-Term Memory* (LSTM) networks and the *Xception* model for stress classification using *electrocardiograms* (ECGs). They extracted heart rate variability features using LSTM networks and converted ECG signals into spectrogram images to capture spatial features using the Xception model.

[23] addresses the challenges of *Unmanned Aerial Vehicle* (UAV) classification in complex environments by proposing a *deep learning fusion approach* that leverages multiple spectrogram representations derived from radar signals. They integrated *micro-Doppler spectrograms*, *cadence-velocity diagrams*, and *cepstrum images* using two fusion strategies including *data-level fusion* and *feature-level fusion*. Their experimental results demonstrate that both fusion strategies significantly outperform methods that use individual spectrograms or raw data alone. Additionally, [24] addresses the challenges of ranking and classifying multidimensional data by introducing two novel methods including *Spanning Thread* (ST) and *Ordered Spanning Thread* (OST). These methods create *linear orderings* that preserve data point similarities, effectively organizing data without altering dimensionality.

3 ClassCube Methodology

In this Section, we present the *ClassCube* methodology for efficient and effective classification of high-dimensional OLAP data cubes by integrating dimensionality reduction techniques with hierarchical data cube structuring. Our approach aims to mitigate computational costs on big OLAP data cube classification while preserving the algorithm's performance.

Consider a big multidimensional OLAP data cube O with N dimensions, denoted as $D = \{d_1, d_2, \dots, d_N\}$. This data cube contains aggregate data across all possible combinations of these dimensions, simplifying complex analytical queries such as slicing, dicing, and drilling down into data. However, performing classification directly on O is computationally expensive due to the high dimensionality.

To address this challenge, we consider a logical cuboid lattice L from O . The lattice L is a hierarchical structure representing all possible aggregations of O over different combinations of dimensions. Each cuboid within L corresponds to a specific level, providing comprehensive multiple views of the data. By working with cuboids of reduced dimensionality, we aim to alleviate the computational costs associated with high-dimensional data.

To provide more details, a cuboid C is an aggregation of the data over a subset $D' \subseteq D$, where D' contains a specified combination of dimensions. The aggregation

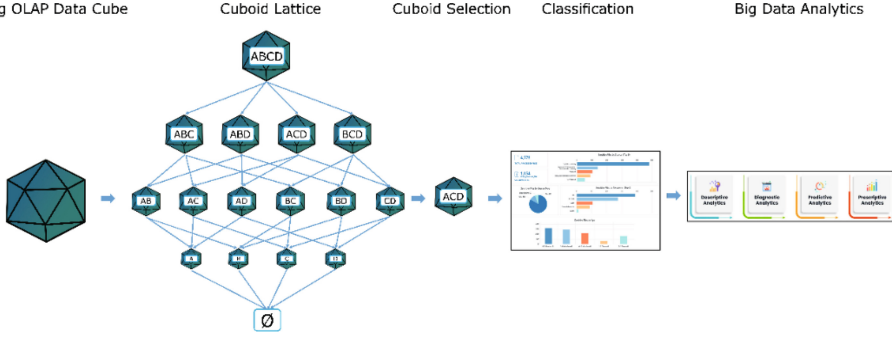


Fig. 1. *ClassCube* Methodology.

within a cuboid may involve operations such as sum, average, or count, over the dimensions in D' . The hierarchical relationship among cuboids is based on the principle of dimension aggregation, where lower-dimensional cuboids are derived by aggregating higher-dimensional ones over one or more dimensions.

Construction of the Lattice Levels: The logical lattice L is defined by hierarchically structured cuboids at different levels as illustrated in Fig. 1. This hierarchical structure enables analysis at various levels of granularity, providing flexibility in the selection of cuboids for classification tasks and addressing challenges related to high dimensionality and computational cost.

To achieve our goal of efficient classification, we propose two approaches to obtain cuboids with the same dimensionality:

- *Dimension Selection;*
- *Principal Component Analysis.*

In both approaches, a specific level k (with dimensionality k_d) of the lattice L is selected using the selection criterion, which is obtained as *user/application input* or determined as previously described based on *state-of-the-art models* (e.g., [13, 14]). Then, a classification algorithm is applied to each cuboid C_{ki} at level k .

Dimension Selection Approach: The dimension selection method involves selecting specific subsets of dimensions from D to create reduced versions of O . For each level k in the lattice L , where $k = 1, 2, \dots, K$ and $K \leq N$, we define m_k as the number of dimensions at level k (i.e., $m_k = k_d$).

At each level k , we select all possible combinations of m_k dimensions from D . Each combination $D_{ki} \subset D$ defines a cuboid C_{ki} , where $i = 1, 2, \dots, I_k$, $I_k = \binom{N}{m_k}$ is the number of combinations at level k . The cuboid C_{ki} is obtained by aggregating O over the dimensions included in D_{ki} .

Therefore, the set of cuboids at level k is as follows:

$$L_k = \{C_{ki} | D_{ki} \subset D, |D_{ki}| = m_k\} \quad (1)$$

Each cuboid C_{ki} has dimensionality m_k and provides a view of the data over m_k ($=k_d$) dimensions. By working with this lower-dimensional cuboid, we reduce the computational complexity of the classification task.

Cuboid Representation Using Principal Component Analysis (PCA). In our second approach, we use PCA to obtain a reduced-dimensionality representation of the data cube O with the same dimensionality k_d as in the dimension selection approach. PCA is applied to O to reduce its dimensionality by transforming the original data into a new set of orthogonal components that capture the maximum variance.

We compute the covariance matrix Σ of O and solve for its eigenvalues and eigenvectors. The top k_d eigenvectors corresponding to the largest eigenvalues form the transformation matrix P . The reduced data cube C_{PCA} is obtained by projecting O onto the new feature space which provides C_{PCA} as follows:

$$C_{PCA} = O \times P \quad (2)$$

This projection results in a single reduced-dimensionality representation, capturing the most significant patterns in the data. By comparing the classification performance using the cuboids from the dimension selection approach and the PCA-reduced data cube, we evaluate the effectiveness of our methodology.

Classification Algorithms: we apply classification algorithms to the selected cuboids S to evaluate the effectiveness of our dimensionality reduction approach. Specifically, we use *Logistic Regression* and *Support Vector Machines*, which are well-suited for handling reduced-dimensionality data.

Logistic Regression models the probability $P(y = 1|x)$ of a binary outcome using the logistic function as follows:

$$P(y = 1|x) = \frac{1}{1 + e^{-(\beta_0 + \beta^T x)}} \quad (3)$$

where β_0 is the intercept, β is the coefficient vector, and x is the feature vector from a cuboid C_{ki} . The parameters β_0 and β are estimated using *Maximum Likelihood Estimation* (MLE). The likelihood function for a set of observations $\{(x_i, y_i)\}$, $i = 1, 2, \dots, n$ is given by:

$$L(\beta_0, \beta) = \prod_{i=1}^n P(y_i|x_i)^{y_i} [1 - P(y_i|x_i)]^{1-y_i} \quad (4)$$

Maximizing the likelihood function (or equivalently, the log-likelihood) obtains estimates of the parameters that best fit the observed data. The optimization is typically performed using iterative algorithms like Newton-Raphson or gradient descent. To further prevent overfitting, regularization methods can be imposed into the LR model. The common regularization terms are as follows:

- *L1 Regularization (Lasso Regression):* Adds a penalty equal to the absolute value of the magnitude of coefficients. It performs feature selection by shrinking some coefficients to zero.
- *L2 Regularization (Ridge Regression):* Adds a penalty equal to the square of the magnitude of coefficients. It prevents large coefficients but does not enforce sparsity.

The regularized cost function becomes as follows:

$$J(\beta_0, \beta) = -\frac{1}{n} \sum_{i=1}^n [y_i \log P(y_i|x_i) + (1 - y_i) \log(1 - P(y_i|x_i))] + \lambda R(\beta) \quad (5)$$

where $R(\beta)$ is the regularization term and λ is the regularization parameter controlling the trade-off between fitting the data and keeping the model coefficients small.

Support Vector Machines are powerful supervised learning models used for classification and regression tasks. They are particularly effective in high-dimensional spaces and are robust against overfitting, especially in cases where the number of dimensions exceeds the number of observations. SVMs are well-suited for our methodology, which involves reduced-dimensionality data cubes derived from high-dimensional OLAP systems.

SVM algorithm seeks an optimal hyperplane that maximally separates data points of different classes. For linearly separable data, the objective is to maximize the margin between the closest points (support vectors) of the two classes. The optimization problem is formulated as:

$$\min_{w,b} \frac{1}{2} \|w\|^2 \quad (6)$$

$$s.t. \ y_i(w^T x_i + b) \geq 1, \forall i$$

where w is the normal vector to the hyperplane, b is the bias term, x_i are the feature vectors, and $y_i \in \{-1, 1\}$ are the class labels. By solving this convex optimization problem, SVM identifies the hyperplane that not only separates the classes but does so with the greatest possible margin, enhancing the model's generalization capabilities.

Real-world data often cannot be perfectly separated by a linear hyperplane. To address this, SVM introduces the concept of kernel functions. Kernel function projects the original feature space into a higher-dimensional space where linear separation is possible. Common kernels include:

- *Linear Kernel*: $K(x_i, x_j) = x_i^T x_j$.
- *Polynomial Kernel*: $K(x_i, x_j) = (\gamma x_i^T x_j + r)^d$.
- *Radial Basis Function (RBF) Kernel*: $K(x_i, x_j) = \exp(-\gamma \|x_j - x_i\|^2)$

Selecting appropriate hyperparameters is crucial for SVM performance: (i) *Regularization Parameter C*: Determines the penalty for misclassification. A large C prioritizes classification accuracy on the training data, potentially at the expense of generalization. (ii) *Kernel Parameters*: Parameters like γ in the RBF kernel or d in the polynomial kernel affect the flexibility of the decision boundary. (iii) *Cross-Validation*: Techniques such as k -fold cross-validation are used to systematically explore combinations of hyperparameters to identify the optimal model settings.

4 Case Study: Virus Classifications in Healthcare Data

In modern healthcare, accurate and on-time identification of viral infections is essential for patient management, epidemiological tracking, and containment efforts. However, healthcare data often consists of large, high-dimensional datasets with multiple features like patient demographics, clinical symptoms, and various biomarkers. Applying traditional classification techniques directly on such large datasets can be computationally expensive and inefficient, especially when rapid responses are required.

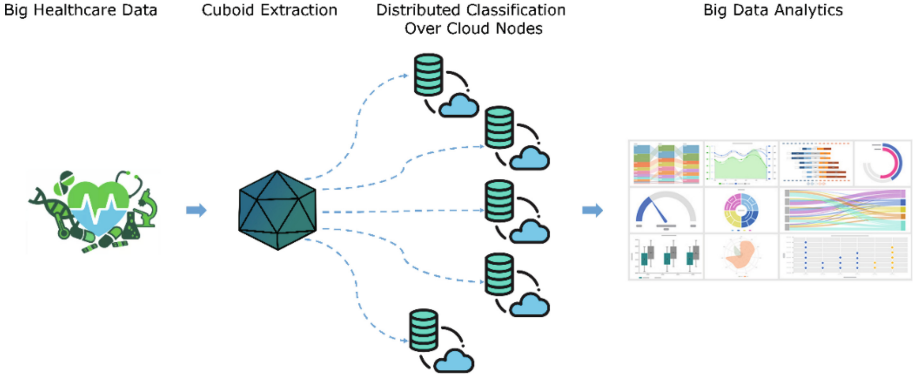


Fig. 2. Case Study: Virus Classification in Healthcare Data.

In this case study, we demonstrate the application of our classification method on a healthcare dataset focused on virus classification, illustrating the method's effectiveness and efficiency through Cloud-based big data analytics. An overview of the architecture of our proposed approach is shown in Fig. 2. In the following, we provide more details about the approach using a healthcare case study.

Healthcare Data Structure: the first stage of our approach involves preparing the high-dimensional healthcare data for classification. In our virus classification example, the dataset consists of numerous dimensions, including patient information (e.g., age, medical history), viral load, and various biomarkers (e.g., antibody levels, protein expressions) that provide distinguishing characteristics among different virus types, such as influenza or coronavirus. The data is structured in an OLAP format, enabling complex multidimensional analysis essential for healthcare applications.

Cuboid Extraction: in this stage, we decompose the high-dimensional OLAP data cube O into multiple lower-dimensional cuboids C_{ki} , each representing an aggregate of the data across specific subsets of dimensions. This process, referred to as cuboid extraction (as illustrated in Fig. 2), uses the hierarchical cuboid lattice L to provide analysis at various levels of granularity.

Formally, let $D = \{d_1, d_2, \dots, d_N\}$ denote the set of all dimensions in the original data cube O . Each cuboid C_{ki} , is defined by a subset of dimensions $D_{ki} \subset D$, where $|D_{ki}| = N - k$ and $k \in \{0, 1, 2, \dots, N\}$ referred to the level number. The total number of

possible cuboids at level k is given by the combination $\binom{N}{N-k}$ representing all combinations of k dimensions from D . Each cuboid C_{ki} is obtained by aggregating the original data cube O over the dimensions not included in D_{ki} . The aggregation operation can be formalized as follows:

$$C_{ki} = AGG(O, \{d \in D | d \notin D_{ki}\}) \quad (7)$$

where AGG represents an aggregation function such as SUM , AVG , or $COUNT$ over the specified dimensions. The set of cuboids at level k is then defined as L_k . By decomposing O into cuboids C_{ki} , we generate a set of smaller, more manageable data cubes that capture specific views of the data. For example, in our healthcare case study, one cuboid might aggregate patient data over demographic dimensions (e.g., *age*, *gender*), while another focuses on specific biomarkers (e.g., *antibody levels*, *protein expressions*).

To evaluate our approach with another dimensionality reduction method, we apply PCA to reduce the dimensionality of the original OLAP data cube O . PCA projects the data onto a lower-dimensional space by retaining the principal components that capture the most significant variance. This process helps in simplifying complex data structures, ensuring that the most informative features are obtained in each cuboid. Therefore, through the cuboid extraction process, whether via dimension selection or PCA, we obtain a set of cuboids $\{C_{ki}\}$ or a reduced data cube C_{PCA} that are suitable for efficient classification and analysis in subsequent stages. By reducing dimensionality, we alleviate the computational load required for classification, making it possible to handle each cuboid independently in a distributed environment.

Distributed Classification over Cloud Nodes: in the third stage, we select a specific cuboid C_{k^*} from the set of cuboids $\{C_{ki}\}$ based on a predefined criterion, namely Φ , and distribute this cuboid over the Cloud nodes for parallelized classification tasks. Consider $C = \{C_{ki} | D_{ki} \subset D, |D_{ki}| = N - k\}$, then this procedure is defined as follows:

$$C_{k^*} = \Phi(C) \quad (8)$$

Once the specific cuboid C_{k^*} is selected, it can be distributed on the Cloud environment for classification tasks, as shown in Fig. 2. Thus, C_{k^*} is partitioned into M subsets to be distributed over the Cloud nodes $v = \{n_1, n_2, \dots, n_M\}$. The partitioning is done as follows:

$$\bigcup_{m=1}^M C_m = C_{k^*}, (C_m \cap C_{m'} = \emptyset, \forall m \neq m') \quad (9)$$

Each Cloud node n_m processes its assigned subset C_m independently, applying classification algorithms such as LR and SVM. For virus classification, LR calculates the probability of each virus type based on the cuboid's features, while SVM identifies a hyperplane that maximally separates the virus classes within the reduced feature space. By operating on reduced-dimensional cuboids, both algorithms are able to perform efficiently without encountering the computational challenges typically associated with high-dimensional data especially when enriched by parallel processing over the Cloud nodes. This approach enables our model to scale with large healthcare datasets, maintaining low latency and enabling real-time analysis.

Big Data Analytics: the final step in our approach, Big Data Analytics, involves aggregating and analyzing the classification results from all Cloud nodes. By combining the output from each node, we obtain comprehensive analytics on virus classification across the entire healthcare dataset. Our approach also leverages multi-resolution analysis enabled by the cuboid lattice, which means that virus classification can be evaluated at different aggregation levels. This flexibility gives us the ability to capture insights at varying levels of detail.

5 Experimental Results

In this Section, we present the results of our experiments conducted on a 6D OLAP data cube built on top of the state-of-art dataset *Adult* [25]. Particularly, we employed two dimensionality reduction techniques, namely Dimension Selection approach, and PCA. We evaluated the classification performance using LR and SVM on the reduced data cubes and compared these results together as well as the performance on the entire OLAP cube. Key performance metrics include *Accuracy*, *Confusion Matrices*, and *ROC curves with AUC values*.

Our first aim was to evaluate the efficacy of reducing the dimensionality of the OLAP data cube from 6D to 4D. We applied two key dimensionality reduction techniques: PCA and Dimension Selection, and employed LR and SVM to classify the data.

Starting with LR, we observe in Fig. 3(a) that the performance on the original 6D data cube is well with AUC values nearing 1.0 for most classes, peaking at 0.99732. The corresponding confusion matrix (see Fig. 3(b)) illustrated a reasonable classification accuracy, demonstrating the model's effectiveness when all six dimensions were utilized.

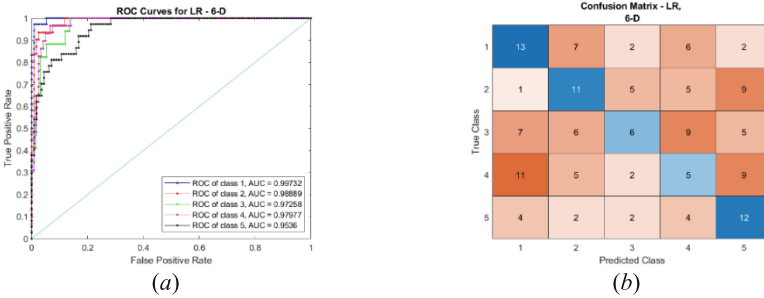


Fig. 3. AUC Curves (a) – Confusion Matrix (b) for Logistic Regression over 6D Data Cube.

Similarly, the SVM classifier shows good performance on the 6D data cube (see Fig. 4), with AUC values slightly lower than those of Logistic Regression, reaching up to 0.97977.

However, when we reduced the data cube to 4D using PCA, the LR classifier continued to deliver good results. As shown in Fig. 5a, the AUC values ranged from 0.93835 to 0.99858, with only a slight decline in performance for some classes. The confusion matrix (see Fig. 5b) reveals a marginal increase in misclassifications, but overall, the

accuracy remained strong. This indicates that PCA successfully captured the most significant features of the data, enabling the 4D model to perform almost as well as the 6D model. The small decrease in classification power is a reasonable trade-off considering the reduced computational complexity, suggesting that the 4D model is a highly efficient alternative.

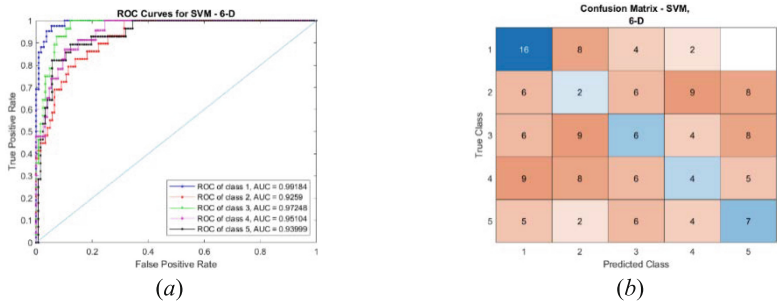


Fig. 4. AUC Curves (a) – Confusion Matrix (b) for SVM over 6D Data Cube.

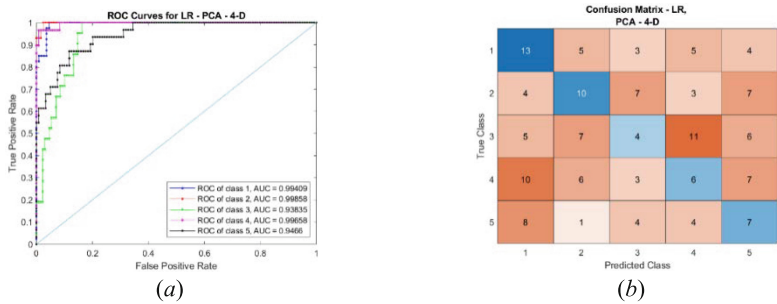


Fig. 5. AUC Curves (a) – Confusion Matrix (b) for Logistic Regression with PCA over 4D Data Cube.

Additionally, the confusion matrix shows that SVM effectively classified most data points, though it struggled with certain overlapping classes. When the dimensionality was reduced to 4D using PCA, the AUC values for SVM ranged from 0.90146 to 0.98541 (see Fig. 6a). Although there was a more noticeable drop in performance for some classes compared to Logistic Regression, the overall classification accuracy remained satisfactory. The confusion matrix (see Fig. 6b) highlighted more significant challenges in class separation, but given the substantial decrease in complexity, the 4D SVM model still offers an efficient solution.

The Dimension Selection approach followed a similar trend. Using Logistic Regression, when the dimensionality was reduced to 4D using Dimension Selection, the performance was comparable to that of PCA. As shown in Fig. 7a, the AUC values were slightly lower, but the overall accuracy remained robust. The confusion matrix (see

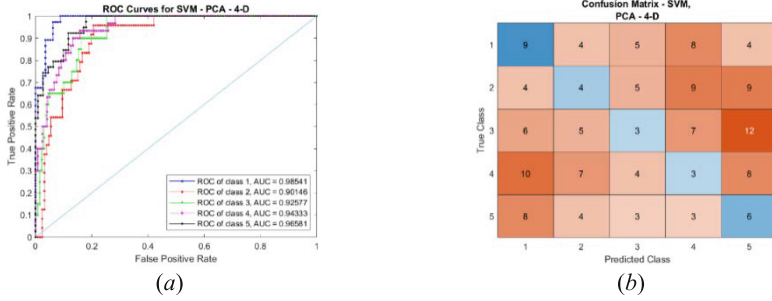


Fig. 6. AUC Curves (a) – Confusion Matrix (b) for SVM with PCA over 4D Data Cube.

Fig. 7b) indicated a few additional misclassifications, but the results were still reasonable, demonstrating that the selected dimensions preserved critical information for classification. This result reinforces the value of the Dimension Selection method, particularly in scenarios where interpretability is crucial, as it provides an understandable set of features without sacrificing much performance.

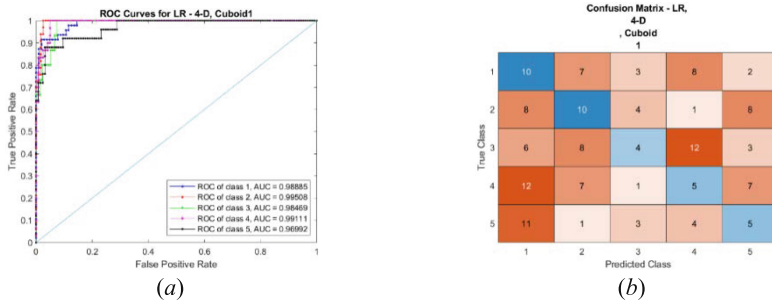


Fig. 7. AUC Curves (a)– Confusion Matrix (b) for Logistic Regression with Dimension Selection over 4D Data Cube.

Regarding the SVM classifier, when combined with Dimension Selection, there was a slight decrease in performance when dimensionality was reduced to 4D. As shown in Fig. 8a, the AUC values ranged from 0.90346 to 0.96861, and the confusion matrix (see Fig. 8b) illustrates increased misclassifications for certain classes.

The comparison between the 4D cuboid and the 6D original OLAP cube shows that dimensionality reduction has minimal impact on classification performance, especially for Logistic Regression. The slight drop in accuracy is acceptable considering the reduced complexity. This demonstrates that the approach effectively mitigates the curse of dimensionality while preserving predictive power and enabling efficient Cloud-based deployment.

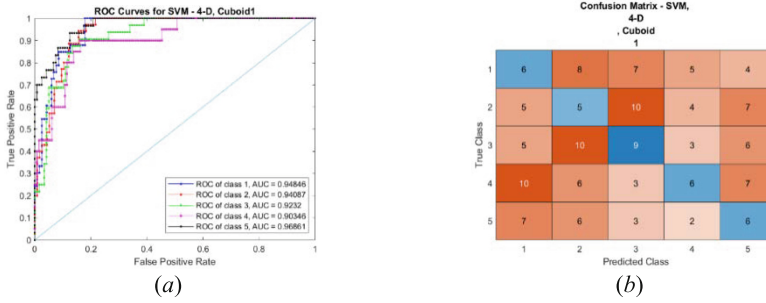


Fig. 8. AUC Curves (a) – Confusion Matrix (b) for SVM with Dimension Selection over 4D Data Cube.

6 Conclusions and Future Work

In this paper, we propose and experimentally assess *ClassCube*, an innovative methodology for effective and efficient big OLAP data cube classification via dimensionality reduction techniques. Our proposed approach leverages *logical cuboid lattices* to represent data at multiple aggregation levels, which enables efficient selection of dimensions and cuboids for classification tasks. The experimental results highlight the trade-off between *reduced computational overhead* and maintained *classification accuracy*. The use of *Logistic Regression* and *SVM* on reduced-dimensional cuboids highlights that our approach effectively preserves performance while significantly reducing resource requirements.

Future work is mainly oriented through extending our methodology with advanced *Machine Learning* models to further enhance *flexibility* and *scalability* as well as integrate emerging *big data trends* (e.g., [26–30]).

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